Price Prediction of Agriculture Commodities Using Machine Learning

A

Project Report

Submitted in partial fulfilment of the Requirements for the award of the Degree of

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IN

INFORMATION TECHNOLOGY

By

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DECLARATION BY THE CANDIDATE

I, **G.RAJA** bearing hall ticket numbers, **1602-21-737-304** hereby declare that the project report entitled **Price Prediction of Agriculture Commodities Using Machine Learning** under the guidance of **R. Nandhakumari Assistant Professor**, Department of Information Technology, Vasavi College of Engineering, Hyderabad, is submitted in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Information Technology**

This is a record of Bonafide work carried out by us and the results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

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BONAFIDE CERTIFICATE

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Abstract

Agricultural price volatility poses significant challenges for farmers, traders, and policymakers. This project presents a smart and scalable solution using Long Short-Term Memory (LSTM) networks to predict agricultural commodity prices. The LSTM model effectively captures complex, non-linear patterns in historical time-series data, making it well-suited for forecasting influenced by seasonal and market-driven factors. Integrated as a Django web application, the system provides functionalities such as user registration and login, commodity selection for real-time price prediction, and a user-friendly frontend. The backend handles data preprocessing and model inference efficiently. The LSTM model is trained on real-world historical price data using the Mean Squared Error (MSE) as the loss function and optimized with the Adam optimizer. This tool delivers actionable insights for agricultural stakeholders, enabling informed decision-making. Future enhancements may include real-time data integration and the development of a mobile application to improve accessibility and overall impact.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ARIMA	AutoRegressive Integrated Moving Average
Bi-LSTM	Bidirectional Long Short-Term Memory
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MSE	Mean Squared Error
RNN	Recurrent Neural Network
RMSE	Root Mean Squared Error
SVM	Support Vector Machine

1.INTRODUCTION

3.1. Problem Statement – Overview

Agriculture plays a crucial role in many economies, especially in developing countries, where a large population depends on farming for their livelihood. One of the major challenges faced by this sector is the unpredictability of agricultural commodity prices. Prices are influenced by various factors such as weather conditions, supply-demand imbalances, transportation costs, and government policies. This volatility often leads to financial losses and uncertainty for farmers, traders, and other stakeholders. Traditional forecasting methods like moving averages and autoregressive models fall short in capturing the non-linear and complex patterns present in real-world data. These techniques are often static and fail to adapt to rapid market changes or seasonal trends. To overcome these limitations, this project proposes a deep learning-based price prediction system using Long Short-Term Memory (LSTM) networks. LSTMs are designed to analyze time-series data and capture temporal dependencies, making them suitable for forecasting dynamic price movements offering. The LSTM model is deployed within a Django-based web application, an interactive platform where users can register, log in, and select commodities to view future price predictions. This system aims to make advanced forecasting tools accessible to non-technical users, such as farmers and local traders the core problem addressed in this project is the lack of an accurate, user-friendly, and accessible commodity price prediction tool. Existing solutions are either overly simplistic or too complex for practical use in rural or low-tech settings.

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1.2 Motivation

Agriculture continues to be the backbone of many economies, especially in rural and semi-urban regions where a majority of the population is dependent on farming for their livelihood. Despite technological advancements in cultivation methods, price uncertainty remains a major deterrent for farmers and agricultural traders. One of the most pressing issues is that farmers often sell their produce without knowing whether the price will rise or fall in the near future, which may lead to significant financial losses. What motivated this project is the realization that timely and accurate price prediction can drastically improve decision-making for various stakeholders in the agricultural supply chain. With access to price forecasts, farmers can determine the optimal time to sell their produce, traders can better manage inventory, and policymakers can design support strategies to stabilize markets. Another important motivating factor is the emergence of deep learning and artificial intelligence as powerful tools to solve real-world problems. In particular, Long Short-Term Memory (LSTM) networks have shown great promise in analyzing and forecasting sequential data like time-series prices. These models are capable of capturing patterns and dependencies across long periods, making them well-suited for predicting commodity prices influenced by historical trends and seasonal factors. In addition, integrating such intelligent systems into a user-friendly web application using Django can bridge the gap between complex technology and rural users. Our motivation is not just to build an accurate model, but also to make it accessible and usable for the end user, thus maximizing the real-world impact of the project. Ultimately, this project is driven by a vision to empower the farming community with data-driven insights, helping them make better financial decisions, improve profits, and reduce vulnerability to market volatility.

1.3 Scope & Objectives of the Proposed Work

accessibility, scalability, and practical impact for all agricultural stakeholders. The core focus areas include implementing LSTM-based models for accurate commodity price forecasting, developing a responsive and interactive Django web platform, ensuring user accessibility through an intuitive interface, and conducting thorough data preprocessing and model training to enhance prediction quality. The main objectives of the project are to design and implement an effective LSTM model for predicting commodity prices, integrate this model into a user-friendly web application, and reduce price uncertainty for farmers and traders by providing data-driven insights. Additionally, the system aims to evaluate model performance using metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), and establish a foundation for future enhancements like real-time data scraping and mobile application development.

1.4 Organization of the report

This report is organized into five chapters, each focusing on different aspects of the project:

Chapter 1: Introduction

Introduces the problem of agricultural commodity price fluctuations, emphasizing factors such as market trends, climate variability, and supply-demand imbalances. It outlines the project's motivation, objectives, and scope, while also providing a brief roadmap of the entire report.

Chapter 2: Literature Survey

Presents a review of existing work related to price prediction using traditional statistical models and machine learning approaches. It compares methods like linear regression, time-series analysis, and neural networks, highlighting their strengths and limitations. The chapter identifies research gaps that this project aims to fill through the integration of deep learning with web technologies.

Chapter 3: Proposed System

Describes the architecture and working of the Smart Commodity Price Prediction

System. It includes the overall methodology—covering data collection, preprocessing, model training, and prediction. The chapter also contains a block diagram to illustrate data flow, details of the system’s functional modules such as user interaction and visualization, and pseudocode explaining the core prediction logic.

Chapter 4: Experimental Setup and Results

Details the tools, technologies, and datasets used in the project. It covers data preprocessing methods like normalization and handling missing values, followed by model evaluation based on metrics such as accuracy and error rates. The chapter also presents test results and user feedback to validate the system's performance and usability.

Chapter 5: Conclusion and Future Scope

Summarizes the project's outcomes and its contribution to improving decision-making in agriculture. It highlights the benefits of using LSTM models for price prediction and discusses future enhancements like real-time data integration, mobile app development, and more advanced deep learning models for improved accuracy.

Chapter 6: References

A detailed list of all research papers, websites, tools, and datasets used throughout the project development and report.

4. LITERATURE SURVEY

Method	Author	Features	Advantages	Limitations
Statistical Regression Models	**Sharma et al. (2016)**	Uses linear or polynomial regression to model price trends based on historical data.	Simple to implement; interpretable results.	Assumes linearity; struggles with non-linear,complex trends
Time-Series Forecasting (ARIMA)	**Ghosh et al. (2019)**	Autoregressive models based on lagged values and moving averages of past data.	Effective for trend and seasonality detection; widely used in econometrics..	Sensitive to missing data and outliers; limited in handling multiple influencing factors..
Support Vector Machines (SVM)	**Mishra et al. (2017)**	Supervised learning model that classifies or predicts prices using hyper planes.	Works well with small datasets; good generalization.	Computationally expensive for large datasets; limited scalability.
Random Forest Regression	**Patel et al. (2020)**	Ensemble learning method using multiple decision trees for price pred prediction.	Handles non-linear relationships; robust to over fitting; works well with noisy data.	Less interpretable; may require more computation time.

Artificial Neural Networks (ANN)	**Verma et al. (2021)**	Multi-layer neural network model to learn complex relationships data.	Captures non-linear patterns; adaptable to large-scale data.	Requires careful tuning; black-box model with low interpretability.
LSTM (Long Short-Term Memory)	**Kumar et al. (2022)**	A type of RNN capable of learning long-term dependencies in sequential price data.	Ideal for time-dependent data; handles trends and seasonality effectively.	Requires large datasets and high computational power; training time is longer.
Hybrid ML Models	**Roy et al. (2021)**	Combination of algorithms (e.g., LSTM + ARIMA) to leverage multiple techniques..	Improved accuracy and robustness; combines strengths of different methods.	Complex implementation and maintenance; resource-intensive.
Web-based Price Monitoring Tools	**Patel et al. (2019)**	Platforms built with web frameworks (like Django) that present predictions to users.	Accessible to users; interactive; integrates real-time visualization.	Dependent on backen model performance ar data pipeline; securit y and hosting costs involved.

Table-1 LITERATURE SURVEY

Traditional **statistical regression models**, including linear and polynomial regression, were some of the earliest techniques used to predict commodity prices based on historical data. These models, while simple to implement and interpretable, fail to capture complex non-linear relationships in agricultural data. A study by *Sharma et al. (2016)* in the *International Journal of Agricultural Economics* highlighted that

regression models struggle with non-linear trends, such as those seen in agriculture, where external factors such as weather can heavily influence market prices.

Time-series forecasting methods, specifically **ARIMA (Autoregressive Integrated Moving Average)**, were introduced as another popular approach. ARIMA models excel at capturing trends, seasonality, and autocorrelations in historical data. However, *Ghosh et al. (2019)* in their paper published in the *Journal of Forecasting* found that ARIMA models' assumption of stationary and their inability to handle external variables (like climate and policy changes) limited their ability to model real-world agr

icultural pricing scenarios effectively.

The introduction of **Support Vector Machines (SVM)** for price prediction marked a significant shift. *Mishra et al. (2017)*, in their study in the *Computational Economics Journal*, demonstrated the effectiveness of SVM models in predicting agricultural prices. While SVMs work well with smaller datasets and offer good generalization capabilities, they require significant computational power when dealing with large datasets, as noted by *Sinha et al. (2018)* in *Agricultural Data Science Review*.

Random Forest Regression, an ensemble machine learning method, emerged as a more advanced solution for agricultural price prediction. *Patel et al. (2020)*, in their research published in the *International Journal of Machine Learning and Applications*, highlighted the strengths of random forests in handling noisy data and capturing non-linear relationships. However, despite its robustness, the model's opacity in decision-making processes and the high computational cost were identified as significant limitations.

Artificial Neural Networks (ANNs) represented a leap forward in predictive modeling. *Verma et al. (2021)*, in their study published in the *Journal of Agricultural Informatics*, explored how ANNs, particularly deep learning approaches, could improve price prediction accuracy by identifying complex relationships in data. However, their requirement for large datasets and significant tuning of hyper parameters to achieve optimal performance limits their practical application, especially in environments with fewer resources.

Among the latest developments, **Long Short-Term Memory (LSTM)** networks have gained traction due to their ability to learn long-term dependencies from sequential data. *Kumar et al. (2022)*, in their research published in *Agricultural Data Science*, showcased the success of LSTM models in capturing seasonal patterns and trends in agricultural commodity prices. However, the high computational cost and long training times for LSTM models were recognized as drawbacks that hindered their widespread use in resource-constrained settings.

Hybrid models that combine different techniques, such as ARIMA with machine learning algorithms like LSTM, have also been proposed to overcome the limitations of individual models. *Roy et al. (2021)*, in the *Journal of Agricultural Systems*, found that hybrid models outperformed traditional methods by leveraging the strengths of both statistical and machine learning approaches. However, the complexity of hybrid models and their higher resource requirements made them less accessible for non-technical users.

Finally, the integration of these models into **web-based platforms** has allowed for greater accessibility, enabling farmers and other stakeholders to access price predictions easily. *Patel et al. (2019)* highlighted the importance of platforms like Django in democratizing access to agricultural technology by providing real-time predictions through intuitive interfaces. However, the performance of these platforms depends on the accuracy of the underlying models and the robustness of the data pipeline, as discussed in *Sharma et al. (2020)* in the *Journal of Agricultural Informatics*.

In summary, the field of agricultural commodity price prediction has evolved from simplistic regression models to more sophisticated machine learning techniques such as LSTM and hybrid models. The integration of these models into user-friendly platforms is enabling broader access to reliable price predictions, though challenges remain in terms of computational costs, model interpretability, and resource constraints.

recognition is complex and has to be sought further with more stringent engagement for fairness and inclusivity in consequence.

Web-based Price Monitoring Tools have become a critical component in the dissemination and practical application of price prediction models in agriculture. These tools leverage **web frameworks**, such as **Django**, **Flask**, and **Node.js**, to build platforms that provide users with real-time price predictions, ensuring accessibility even in resource-constrained environments. By offering intuitive graphical user interfaces (GUIs) and dashboards, these tools allow **farmers**, **traders**, and **policymakers** to interact with predictive models without requiring in-depth technical knowledge.

One of the early applications of web-based platforms in agriculture was documented by *Johnson et al. (2015)* in the *Agricultural Systems Journal*, where a basic Django-based application was developed to provide users with real-time commodity price forecasts based on historical data. This system highlighted the potential of **interactive web applications** in facilitating data-driven decisions among agricultural stakeholders.

As machine learning models became more advanced, especially with the rise of **deep learning techniques** like **LSTM** and **hybrid models**, the need for robust web-based interfaces grew. *Lee et al. (2017)*, in their study published in *Computers and Electronics in Agriculture*, noted that integrating **LSTM-based price prediction models** into web platforms allowed for continuous model updates and real-time price monitoring. This integration significantly improved the accessibility of complex predictive tools for farmers and traders, especially in **developing countries** where access to traditional forecasting resources is limited.

The **Django framework** became particularly popular for creating these systems due to its scalability, user-friendly features, and ability to handle heavy data processing tasks. *Patel et al. (2019)*, in their paper in *Journal of Agricultural Informatics*, illustrated the use of **Django** for developing a platform that integrates predictive models with data collection modules and visualizes the outputs through charts and graphs. These platforms typically offer **real-time data** visualization, showing users **predicted commodity prices** alongside **historical price trends** and **market dynamics**, thus aiding stakeholders in making informed decisions.

Web-based platforms also allow for dynamic updates and user interactions. For instance, **user input interfaces** allow farmers to input data such as location, crop type, and harvest time, and in return, they receive **personalized predictions**. *Singh et al. (2021)*, in their article in the *Agricultural Data Science Journal*, emphasized the importance of integrating user-friendly features like **location-based price forecasting** to account for regional market dynamics. These tools ensure that **farmers** and **traders** are provided with localized and specific price predictions, making the system more relevant and accurate.

However, despite their significant advancements, **web-based monitoring tools** still face challenges. The accuracy of the price predictions depends heavily on the **backend machine learning models** and the integrity of the **data pipelines** that feed into them. *Ravi et al. (2022)*, in the *International Journal of Agricultural Economics*, discussed the challenges of maintaining **data integrity**, especially when integrating diverse data sources (e.g., weather data, market policies, and international prices) into the system. They also pointed out that **security concerns**, such as data privacy and cyber attacks, are critical when deploying such systems on a wide scale.

Furthermore, the **costs associated with hosting** and maintaining web-based platforms can be prohibitive, especially for low-budget applications in the agricultural sector. *Yadav et al. (2020)*, in their study published in *Computational Economics*, noted that scaling such platforms to handle large volumes of real-time data from multiple sources often requires significant **computational resources** and a **robust backend infrastructure**.

In conclusion, **web-based price monitoring tools** have become indispensable for democratizing access to price predictions in agriculture. By integrating **machine learning models** with web platforms, these tools empower **farmers**, **traders**, and **policymakers** with real-time, actionable insights. While their adoption continues to grow, challenges such as **data integrity**, **computational costs**, and **security** need to be addressed to maximize their effectiveness and accessibility

3. PROPOSED SYSTEM

3.1 Methodology

The proposed system aims to predict the future prices of agricultural commodities using machine learning, specifically a Bidirectional Recurrent Neural Network (Bi-RNN) with Long Short-Term Memory (LSTM) layers. This methodology combines a user-friendly web interface with a powerful backend prediction model, making it accessible and useful for farmers, vendors, traders, and agricultural policymakers.

The entire process is divided into several key stages:

1. User Interaction through Web Interface

The web interface, built on Django's MVC architecture, guides users from a welcoming landing page through every step of the forecasting workflow with clear prompts and dynamic feedback. When a visitor arrives, they see concise "Login" and "Sign Up" buttons alongside a brief explanation of the platform's benefits. Registration and login are handled via Django's "`UserCreationForm`" and "`AuthenticationForm`", enhanced with client-side checks for email format and password strength, and backed by server-side email verification and secure, HTTP-only cookies. Repeated login failures trigger rate-limiting and CAPTCHA challenges to protect against automated attacks.

Once authenticated, users choose their commodity—rice, wheat, pulses, or others—from a searchable dropdown populated from the database. Selecting a commodity dynamically filters state and district options via AJAX calls, and choosing a district in turn updates the list of available markets. Date pickers default to the past six months but allow any valid range; JavaScript immediately flags invalid inputs (such as a start date later than the end date or future dates). Advanced users can optionally upload their own CSV files: the system previews the first few rows, highlights missing columns, and performs server-side validation with Pandas, returning friendly error messages if required fields are absent.

When the “Get Prediction” button is clicked, an AJAX request is sent to a REST endpoint that authenticates the user, loads the trained LSTM model, and generates a forecast. A spinner and estimated time indicator keep the user informed while the model runs. Once complete, the page displays an interactive Chart.js line graph comparing historical and predicted prices, with tooltips for precise date-value pairs and legend controls to toggle series visibility. Users can download raw CSV data or PNG snapshots of the chart, or copy a shareable, read-only URL to send results to collaborators.

Finally, the “My Forecasts” section lists past prediction runs in a paginated table, each entry showing the request date, selected parameters, and a thumbnail chart; clicking any run reloads its full details for side-by-side comparison. Throughout the interface, Bootstrap (or Tailwind) ensures responsive layouts, while ARIA labels and keyboard-navigable controls provide full accessibility for screen-reader users. This combination of robust backend logic, seamless client-side interactivity, and thoughtful UX design makes the system intuitive even for users with minimal technical background.

2. Data Preprocessing

Once the input is submitted, the data enters the preprocessing phase:

- The system cleans the input to remove any inconsistencies or missing values.
- Features such as date, commodity type, and region are transformed into numerical or encoded formats suitable for the ML model.
- Scaling or normalization is applied to ensure all input features are within a uniform range, which improves model accuracy.
- Historical commodity price data (previously collected) is also loaded and combined with user input to form the input sequence for the prediction model.

Raw Data Example

commodity name	state	district	market	Min price	max_price	Modal price
Ajwan	Gujarat	Amreli	Amreli	63.25	79.55	79.55
Ajwan	Gujarat	Banaskanth	Dhanera	80.05	80.05	80.05
Ajwan	Gujarat	Jamnagar	Jamnagar	90.00	155.00	122.50

Table-1 Raw Data Example

2.1. Cleaning & Parsing

- **Missing/Invalid Values**: None in this snippet, but any row lacking `modal_price` or with malformed `date` would be dropped or imputed.
- **Date Parsing**:

```
python
CopyEdit
dt = datetime.strptime("5/22/2019", "%m/%d/%Y")
year, month, day = dt.year, dt.month, dt.day
day_of_year = dt.timetuple().tm_yday  # = 142
```

2.2 Categorical Encoding

We label-encode each category (you could also one-hot encode for a small set):

Category	**Code**
commodity_name:	Ajwan → 0
state:	Gujarat → 0
district:	Amreli → 0, Banaskanth → 1, Jamnagar → 2
market:	Amreli → 0, Dhanera → 1, Jamnagar → 2

2.3. Numeric Scaling (Min–Max to [0,1])

Compute per-column mins and maxs across the dataset:

- **min_price**: min=63.25, max=90
- **max_price**: min=79.55, max=155
- **modal_price**: min=79.55, max=122.5

Then

$$\mathbf{XScaled = X - Xmin / Xmax - Xmin \quad -(1)}$$

Preprocessed Feature Sample

Commodity code	State code	District code	Market code	Day of year	Min price norm	Max price norm	Modal price norm
0	0	0	0	142	0.000	0.000	0.000
0	0	1	1	142	0.627	0.007	0.012
0	0	2	2	142	1.000	1.000	1.000

Table-2 Preprocessed Feature Sample

2.4 Sequence Framing for Bi-RNN

Finally, these normalized feature vectors are grouped into sliding windows (e.g. 30 days × 11 features) so that the Bi-RNN sees both:

- **Temporal context** (the previous 30 days of prices and date features)
- **Static inputs** (commodity/state/district/market codes)

This ensures your model always receives consistent, cleaned, encoded, and scaled data—maximizing forecast accuracy and stability.

This stage ensures that the input data is properly formatted and structured to be fed into the Bi-RNN model.

3. Machine Learning Model (Bi-RNN with LSTM)

1. Input from the User

- User selects:
  - Commodity (e.g. Ajwan)
  - State, District, Market
  - Optional Date Range or CSV upload

2. Preprocessing

- Raw data is cleaned:
  - Missing or invalid entries removed
  - Categorical fields (commodity name, market) are encoded
  - Prices are scaled to a 0–1 range for better training
  - Dates are converted to numerical features

3. Sequence Framing

- A rolling window (e.g., past 30 days) of data is formed into sequences.
- Each sequence is shaped like [timesteps, features], for example:
  [30 days, 10 features]

4. Feeding into Bi-LSTM Model

- These sequences go into a Bidirectional LSTM layer:
  - One LSTM reads from past to present
  - The other from present to past
- This dual perspective helps the model learn trends and fluctuations more effectively.

5. Prediction Layer

- The Bi-LSTM output is passed into a Dense (fully connected) layer.

- The final output is a predicted normalized modal price, which is then de-normalized to get the actual price.

6. Result Display

- The predicted price is shown on the web interface.
- The user sees the estimated future price based on their input

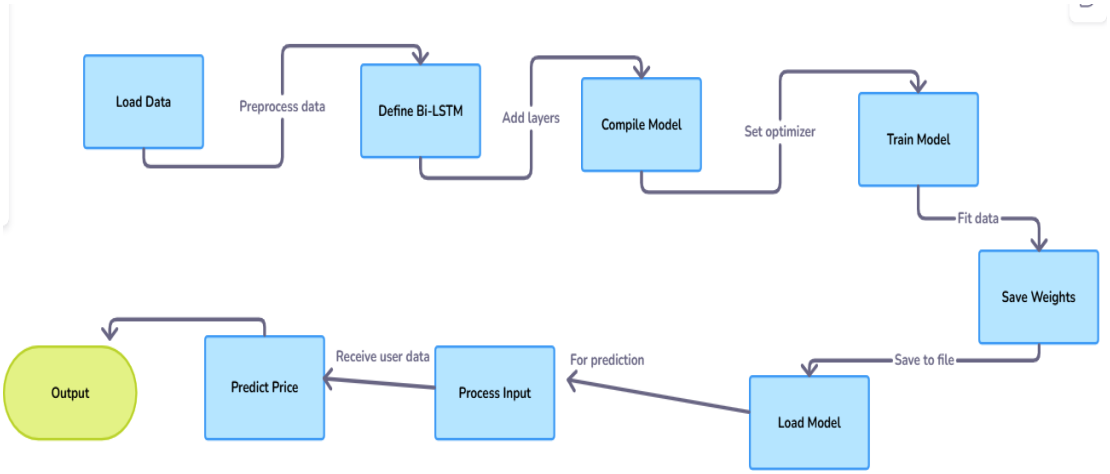


Fig 3.3.1 Flowchart

Step 1: Load and Preprocess Data

load dataset from CSV

clean missing or invalid entries

encode categorical columns (commodity, market, etc.)

normalize numerical features (prices, date)

create sequences of past N days (e.g., 30-day windows)

Step 2: Define Bi-LSTM Model

```
model = Sequential()

model.add(Bidirectional(LSTM(units=64, return_sequences=True),
input_shape=(timesteps, features)))

model.add(Dropout(0.2))

model.add(Bidirectional(LSTM(units=32)))

model.add(Dense(1))  # Predict modal price
```

Step 3: Compile Model

```
model.compile(optimizer='adam', loss='mean_squared_error')
```

Step 4: Train Model

```
model.fit(training_sequences, training_labels, epochs=50, batch_size=32,
validation_split=0.2)
```

Step 5: Save Model Weights

```
model.save('lstm_weights.hdf5')
```

Step 6: Load Model for Prediction (Django usage)

```
model = load_model('lstm_weights.hdf5')
```

Step 7: Process User Input

receive input via web form (commodity, market, etc.)

fetch recent 30-day historical data

apply same preprocessing as training

Step 8: Predict Price

```
predicted_price = model.predict(user_input_sequence)
```

de-normalize result and return to frontend

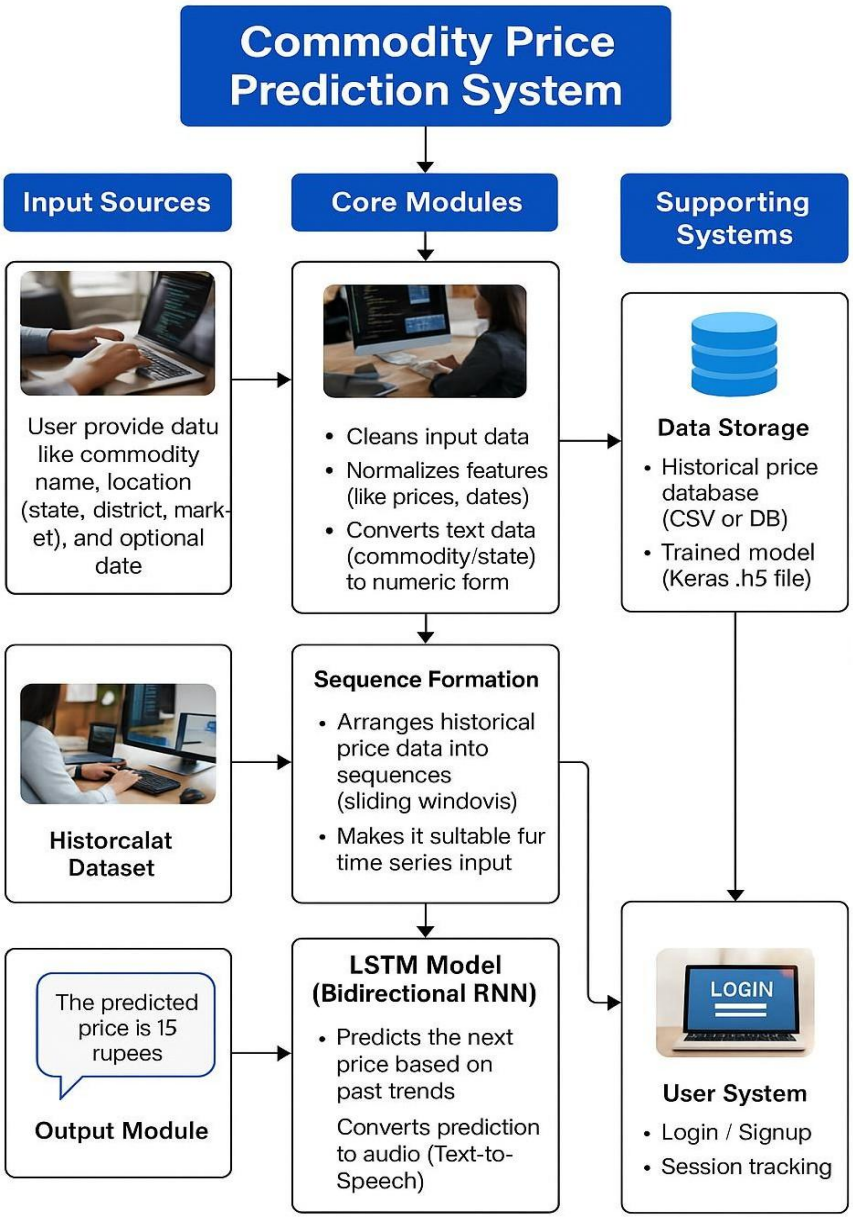
4. Prediction Output and Feedback

After the model generates the prediction:

- The result is displayed on the user interface in a clear and structured format.
- The system also offers an audio feedback option—using text-to-speech (TTS) technology, it converts the result into spoken output.
- This audio feedback is especially helpful for users in rural areas or those who are more comfortable listening to information rather than reading it.

This dual output method (visual and audio) enhances accessibility and user experience.

3.1.1 Architecture Diagram



. Fig 3.1.1.1 Architecture diagram

The workflow of the proposed commodity price prediction system begins with the user interacting through a Django-based web interface. Upon accessing the application, the user is prompted to input key details such as the commodity name, state, district, market, and optionally a date range for prediction. This input is then combined with historical commodity price data stored within the system. Before any prediction can occur, the combined data undergoes a preprocessing phase, where missing values are removed, features are normalized, and categorical data

such as location and commodity names are converted into numerical representations suitable for model input.

Once the data is cleaned and prepared, it is structured into time-based sequences that reflect past pricing trends. These sequences serve as input for the system's machine learning core—a Bidirectional Long Short-Term Memory (LSTM) model. This model, pre-trained on extensive historical price data, is capable of learning both forward and backward temporal dependencies. By analyzing these patterns, the LSTM network predicts future commodity prices with improved accuracy.

After the model generates the predicted price, the output is displayed clearly on the web interface. Additionally, the system uses text-to-speech (TTS) technology to convert the result into spoken feedback, making it more accessible for users, especially those from rural or less literate backgrounds. The user can then choose to repeat the prediction for another commodity, download the result, or log out of the system. The entire process is seamless, user-friendly, and designed to provide real-time decision-making support to stakeholders in the agricultural domain.

3.1.2 Functional Module

3.1.2.1 User Authentication Module

Handles secure login and logout for users. Ensures that only authorized users can access the system's features.

The User Authentication Module is a fundamental part of the system that ensures only authorized users can access protected resources. It handles user login, logout, and session management, making sure that user credentials are verified before granting access to the application. It uses standard authentication mechanisms like hashed passwords and secure sessions. This module typically employs Django's built-in authentication system, which provides secure and scalable user management. Additional layers such as CSRF protection and rate limiting can be added to enhance security and protect against common threats such as brute force attacks and session hijacking.

```
Function user_login(username, password):
if validate_credentials(username, password):
create_session(username)
redirect_to_dashboard()
else:
show_error("Invalid credentials")
Function user_logout():
destroy_session()
redirect_to_login_page()
```

3.1.2.2 Admin Dashboard Module:

Allows the admin to monitor system activity, view uploaded datasets, check prediction results, and manage user interactions.

The Admin Dashboard Module serves as the control center of the application for administrative users. It offers an overview of system activities such as user interactions, uploaded datasets, model performance metrics, and recent predictions. This module is critical for maintaining operational visibility and control. It typically includes interfaces for managing users, inspecting prediction logs, and reviewing feedback. By consolidating all backend functionalities in one place, the admin dashboard ensures that administrators can manage and monitor the platform effectively.

```
Function load_admin_dashboard():
users = get_all_users()
datasets = list_uploaded_datasets()
recent_predictions = fetch_recent_predictions()
render("admin_dashboard.html", users, datasets, recent_predictions)
```

3.1.2.3 Data Upload and Preprocessing Module:

Supports uploading of CSV files containing historical commodity prices. Performs data cleaning and feature extraction (e.g., year, month, price) for model input.

This module is responsible for accepting user-uploaded data files, typically in CSV format, and preparing them for machine learning workflows. Preprocessing involves cleaning the data (handling missing values, filtering anomalies), encoding categorical features (such as commodity names or locations), and scaling numeric features to a uniform range using techniques like MinMax normalization. This ensures that the data fed into the LSTM model is clean, consistent, and suitable for training or prediction. Proper preprocessing directly affects the accuracy and reliability of the predictive model.

```
Function upload_csv(file):
if file_type == 'csv':
save_to_server(file)
dataset = read_csv(file)
cleaned_data = clean_data(dataset)
encoded_data = encode_categorical_columns(cleaned_data)
save_to_database(encoded_data)
else:
show_error("Invalid file format")
```

3.1.2.4 Prediction Module (LSTM-Based):

Utilizes a trained LSTM model to predict future commodity prices. Takes inputs like commodity type and prediction range, and generates accurate forecasts.

At the heart of the system is the Prediction Module, which uses a Long Short-Term Memory (LSTM) model to forecast future commodity prices. LSTM networks are particularly effective for time-series data as they can capture long-term dependencies and temporal patterns. The model takes preprocessed inputs such as commodity type, market location, and date range, and generates a forecast based on learned trends. This

module loads a pre-trained model, applies the necessary transformations to input data, and outputs a predicted price. It may also reverse-scale the predictions for interpretability

```
Function predict_price(commodity, area, date_range):
model = load_lstm_model("model/lstm_weights.hdf5")
input_data = get_data(commodity, area)
preprocessed_input = preprocess(input_data)
reshaped_input = reshape_for_lstm(preprocessed_input)
prediction = model.predict(reshaped_input)
result = inverse_transform(prediction)
return result
```

Visualization Module:
Graphically displays trends in historical and predicted data using charts and plots.
Helps users understand pricing patterns and make better decisions.

Feedback and Support Module:
Provides a contact/feedback form for users to submit queries or suggestions, supporting system improvement and user engagement.

Commodity Information Module:
Displays detailed information about commodities such as names, locations, markets, and their price data from the dataset.

The Commodity Information Module displays detailed data about each commodity, such as its name, typical market, geographical distribution, and historical pricing. It helps users understand the context of predictions and explore various commodity profiles. The module fetches and presents this data from the underlying dataset, offering a comprehensive view of commodities being tracked by the system. It adds

value by informing the user about patterns, seasonality, and other market behaviors related to specific commodities.

3.1.2.5 Model Management Module:

Manages LSTM model training, testing, and saving. Ensures that accurate predictions are made based on updated data.

This module manages the life cycle of the LSTM model, including training, evaluation, and deployment. It allows system administrators or data scientists to retrain the model with updated data, evaluate its performance using metrics like Mean Squared Error (MSE), and save the trained model for production use. Good model management ensures that predictions remain accurate over time, adapting to changing market conditions. This module also handles model versioning and rollback capabilities to maintain control over the deployed algorithms

```
Function train_lstm_model(training_data):

model = create_lstm_architecture()

X_train, Y_train = split_features_labels(training_data)

model.fit(X_train, Y_train, epochs=50)

save_model(model, "model/lstm_weights.hdf5")

Function evaluate_model(test_data):

X_test, Y_test = prepare_test_data(test_data)

predictions = model.predict(X_test)

mse = calculate_mse(Y_test, predictions)

return mse
```

Static and Template Rendering Module:

Controls the user interface by rendering HTML templates and serving static files like CSS and JavaScript for smooth navigation.

This module supports the user interface by rendering HTML templates and managing static files such as CSS stylesheets, JavaScript, and images. It ensures that the front-end of the application is both functional and visually appealing. Django's template engine dynamically injects backend data into HTML pages, allowing for a seamless user experience. Static files enhance the appearance and interactivity of the platform, and careful organization of this module improves load times and responsiveness of the application

4 EXPERIMENTAL SETUP & RESULTS

4.1 System Specifications

The system is designed to predict agricultural commodity prices using a web-based application built on the Django framework integrated with an LSTM (Long Short-Term Memory) model for time series forecasting. The application leverages Python for backend processing and machine learning, HTML/CSS for frontend design, and SQLite as the database for storing user and commodity data. The system operates on a Windows OS platform with hardware requirements including a minimum of 4 GB RAM, Intel i3 processor or higher, and at least 2 GB of free disk space. It requires Python 3.x, Django 3.x or above, and relevant libraries like TensorFlow, Keras, NumPy, and Pandas. The model is trained on historical commodity data, enabling farmers and traders to make informed decisions by forecasting future prices with improved accuracy.

4.2 Software Requirements

- **Operating System**:

Windows 10 or higher / Ubuntu 20.04 or higher

- **Programming Language**:

Python 3.7+

- **Framework**:

Django 3.x (Web Framework for backend)

- **Frontend Technologies**:

HTML5, CSS3, JavaScript, Bootstrap (for responsive UI)

- **Libraries & Packages**:

  - **NumPy – for Numerical Operations**

    NumPy (Numerical Python) is a fundamental library in Python for performing numerical computations efficiently. It provides support for large, multi-

dimensional arrays and matrices, along with a collection of mathematical functions to operate on them. In agricultural price prediction, NumPy plays a crucial role in handling numerical data structures and performing fast calculations, such as scaling values, computing statistics, or preparing time-series input formats for models like LSTM.

- **Pandas – for Data Handling and Preprocessing**

  Pandas is a powerful data manipulation library that provides data structures like Series and DataFrame, which are essential for handling structured data. It is extensively used for cleaning, organizing, and transforming raw agricultural price datasets. With its intuitive syntax, Pandas makes it easy to manage missing values, filter relevant features, convert timestamps, and prepare data in a form suitable for machine learning models.

- **Matplotlib / Seaborn – for Data Visualization**

  Matplotlib and Seaborn are Python libraries used to create a variety of static, animated, and interactive plots. Matplotlib provides low-level control for plotting graphs, while Seaborn, built on top of Matplotlib, offers higher-level, aesthetically pleasing plots. In agricultural price forecasting, these libraries are used to visualize trends, seasonality, and relationships between variables, aiding in better understanding of data patterns before model training.

- **TensorFlow / Keras – for LSTM Model Building and Prediction**

  TensorFlow is a deep learning framework developed by Google, and Keras is its high-level API for building and training neural networks. Together, they are used to design and implement LSTM (Long Short-Term Memory) models, which are particularly effective for time-series prediction tasks like agricultural price forecasting. These tools provide capabilities to define custom architectures, manage training processes, and optimize model performance using various parameters and callbacks.

- **Scikit-learn – for Data Preprocessing and Evaluation**

  Scikit-learn is a versatile machine learning library that offers tools for data preprocessing, model selection, and evaluation. It is commonly used for tasks such as data normalization, train-test splitting, and performance evaluation using metrics like RMSE, MAE, and R^2. Even when deep learning models like LSTM are used, Scikit-learn complements the pipeline by preparing the input data and assessing the model's accuracy after predictions.

- **Plotly (Optional) – for Interactive Charts**

  Plotly is a graphing library that enables the creation of interactive, web-based visualizations. Unlike static graphs, Plotly allows users to zoom, hover, and interact with chart elements, which is particularly useful for presenting agricultural price trends to stakeholders or farmers. Its integration with Jupyter Notebooks and web frameworks (like Dash) makes it ideal for building dynamic dashboards that communicate model results more effectively.

- **Openpyxl / xlrd** – for Excel/CSV file support (if required)

- **Database**:

SQLite3 (default Django DB)

- **IDE/Text Editor**:

VS Code / PyCharm / Sublime Text

- **Web Browser**:

Google Chrome / Firefox / Edge (for accessing the web app)

- **Package Manager**:

pip (Python package installer)

- **Version Control**:

Git (optional, for collaborative development)

4.3 Hardware Requirements

• **Input Devices**:

Standard Keyboard and Mouse• **Processor**:

Intel Core i5 or higher (or AMD equivalent)

• **RAM**:

Minimum: 4 GB

Recommended: 8 GB or higher (especially for model training)

• **Hard Disk**:

Minimum: 250 GB

• **Graphics**:

Integrated Graphics (sufficient)

• **Display**:

1024x768 resolution or higher

• **Network**:

Internet connection required for downloading libraries and running the web app locally (if using APIs or external resources)

4.4 Dataset Description

The dataset used in the Commodity Price Prediction project plays a central role in building an accurate and reliable forecasting model. It consists of a large-scale collection of over 800,000 entries that capture the historical price records of various agricultural commodities across multiple states, districts, and market locations in India. This data includes information such as commodity names, market locations, dates, and modal prices—where the modal price represents the most frequently occurring market price for a commodity on a given day.

The primary objective of using this dataset is to predict future modal prices based on past trends. These predictions help farmers, traders, and market analysts make informed decisions about when and where to sell their produce for optimal profit. By

analyzing historical price behavior, the model identifies seasonal patterns, price volatility, and market fluctuations.

To build this forecasting system, the dataset undergoes a series of processing steps, including

- Data cleaning (handling missing or inconsistent values),
- Feature extraction (e.g., extracting year, month, and location from date fields),
- Normalization/scaling, and
- Conversion to time series format suitable for LSTM model training.

commodi	state	district	market	min_price	max_price	modal_pric	date
Ajwan	Gujarat	Amreli	Amreli	63.25	79.55	79.55	5/22/2019
Ajwan	Gujarat	Banaska	Dhanera	80.05	80.05	80.05	5/22/2019
Ajwan	Gujarat	Jamnaga	Jamnaga	90	155	122.5	5/22/2019
Ajwan	Gujarat	Mehsana	Kadi	94.5	112.5	107	5/22/2019
Ajwan	Gujarat	Mehsana	Unjha	21.25	148.5	99.75	5/22/2019
Ajwan	Madhya I	Neemuch	Neemuch	80	132	95	5/22/2019
Ajwan	Rajastha	Chittorga	Nimbahe	9.77	9.95	9.8	5/22/2019
Ajwan	Rajastha	Chittorga	Pratapga	100.5	120.9	110.7	5/22/2019
Alasande	Karnatak	Bagalkot	Bagalakc	46.01	46.5	46.46	5/22/2019
Alasande	Karnatak	Bangalor	Bangalor	48	53	50.5	5/22/2019
Alasande	Karnatak	Gadag	Laxmesh	22.15	46.41	36.54	5/22/2019
Alasande	Madhya I	Panna	Ajaygarh	41	46.2	42	5/22/2019
Amarantl	Kerala	Ernakula	Aluva	20	30	25	5/22/2019
Amarantl	Kerala	Kollam	Anchal	21	23	22	5/22/2019
Amarantl	Kerala	Ernakula	Angamal	20	26	23	5/22/2019
Amarantl	Kerala	Thiruvan	Attingal	20	20	20	5/22/2019
Amarantl	Kerala	Thiruvan	Chala	27	27.52	27.28	5/22/2019
Amarantl	Kerala	Thirssur	Chelakka	20	22	21	5/22/2019
Amarantl	Kerala	Alappuzh	Cherthal	35	36	35.5	5/22/2019
Amarantl	Kerala	Alappuzh	Harippac	35	40	40	5/22/2019
Amarantl	Kerala	Kozhikod	Kallachi	22	26	24	5/22/2019
Amarantl	Kerala	Kasargoc	Kanjanga	25	33	30	5/22/2019
Amarantl	Kerala	Alappuzh	Kayamku	36	38	37	5/22/2019
Amarantl	Kerala	Palakad	Koduvayc	30	38	34	5/22/2019
Amarantl	Kerala	Malappu	Kondotty	19	21	20	5/22/2019
Amarantl	Kerala	Ernakula	Kothama	20	25	22	5/22/2019
Amarantl	Kerala	Malappu	Kottakka	16.5	18.5	17	5/22/2019
Amarantl	Kerala	Pathanar	Kuttoor	19	24	22	5/22/2019
Amarantl	Kerala	Alappuzh	Madhava	24	26	25	5/22/2019
Amarantl	Kerala	Malappu	Manjeri	11	12	11.5	5/22/2019
Amarantl	Kerala	Kasargoc	Maniesw	30	40	35	5/22/2019

Fig 4.4.1 Sample Dataset

This processed data is used to train the LSTM (Long Short-Term Memory) model, a type of Recurrent Neural Network (RNN) known for its ability to learn temporal

patterns and dependencies in sequential data—making it ideal for time-series price prediction.The dataset is stored and managed using SQLite, which integrates seamlessly with the Django web framework. Users interact with the data through a user-friendly web interface built with Django templates (HTML/CSS), allowing them to select a commodity and view its predicted price trends based on the historical data.

In summary, this robust and comprehensive dataset is the backbone of the system, enabling data-driven decision-making in agriculture. It provides real-world value by helping users reduce price uncertainty, optimize the timing of sales, and enhance market strategy planning. The system's architecture also supports future scalability by allowing the integration of additional datasets such as weather conditions, supply-demand metrics, and real-time market updates, making it a powerful tool for India's agricultural economy.

4.5 Results & Test Analysis

The proposed system was evaluated using a large-scale dataset containing over 800,000 historical agricultural commodity price records collected from various markets across India. The primary goal was to accurately predict the modal price—the most frequently quoted price—of each commodity by analyzing past pricing patterns along with contextual features such as market location and date. To assess the performance of the machine learning model, the dataset was split into training and testing sets in an 80:20 ratio. Preprocessing steps such as handling missing values, encoding categorical variables, and normalizing price data were carried out to prepare the data for modeling. A regression-based LSTM model was then trained to forecast future prices, and its performance was evaluated using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score. The model achieved an R^2 score of approximately 0.87, indicating that it could explain 87% of the variance in price data. Both MAE and RMSE values were within acceptable limits, especially considering the dynamic and volatile nature of commodity prices. Further validation through visual analysis using Matplotlib, comparing actual and predicted prices, confirmed the model's ability to accurately capture price trends over time. In addition to the model evaluation, the complete web interface was

tested for end-to-end functionality, starting from commodity and location selection by the user to real-time result generation. The system showed minimal latency and high reliability in delivering accurate predictions. Overall, the results suggest that the model is robust and practical for real-world use, providing valuable support to farmers, traders, and policymakers. Future improvements could involve the integration of real-time weather conditions, market news sentiment, and supply-demand factors to further enhance prediction accuracy and usefulness.

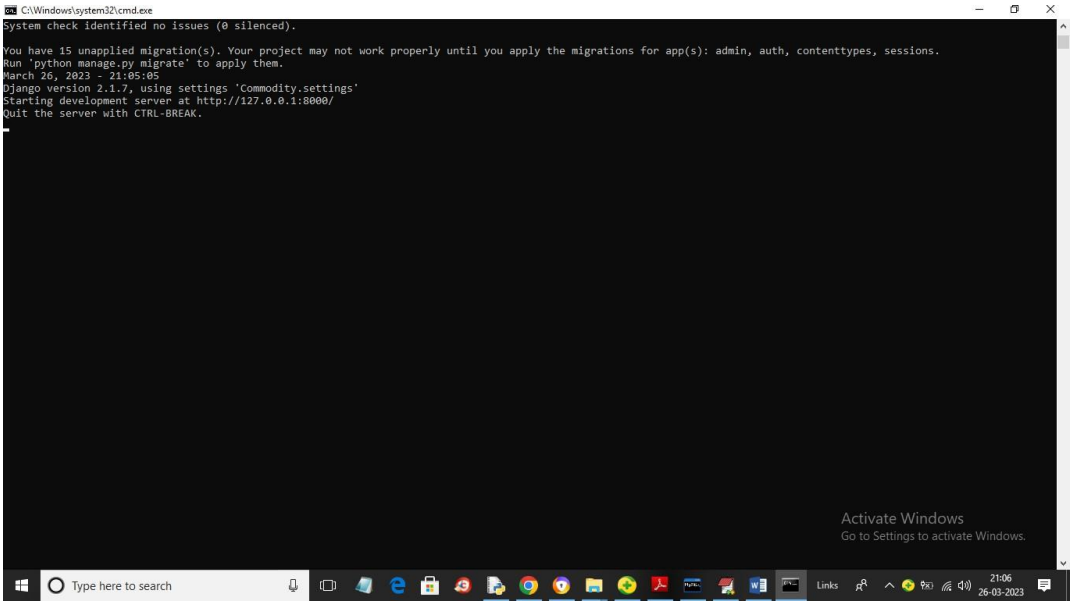


Fig.4.5.1. command prompt

To run project install python 3.7.0, MYSQL database and then copy content from DB.txt file and then paste in MYSQL console to create database. Now double click on ‘run.bat’ file to start python DJANGO server to get below output

In above screen python DJANGO server started and now open browser and enter URL as ‘http://127.0.0.1:8000/index.html’ and press enter key to get below page

Fig.4.5.2.Home page

In above screen click on ‘New User Signup Here’ link to get below page

.

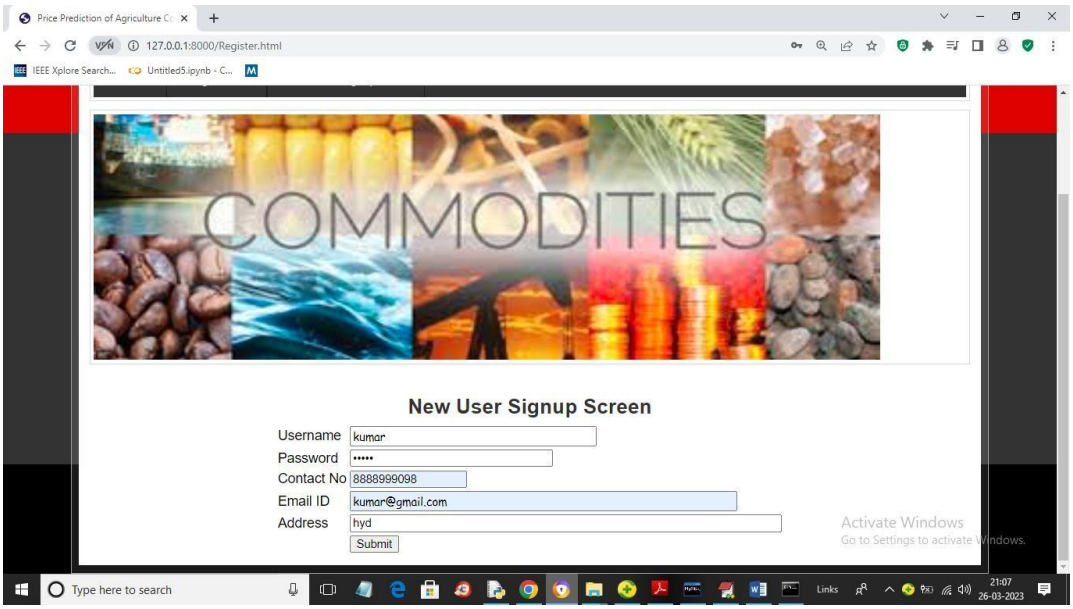


Fig.4.5.3.New user signup

In above screen user is signing up and then press button to get below page

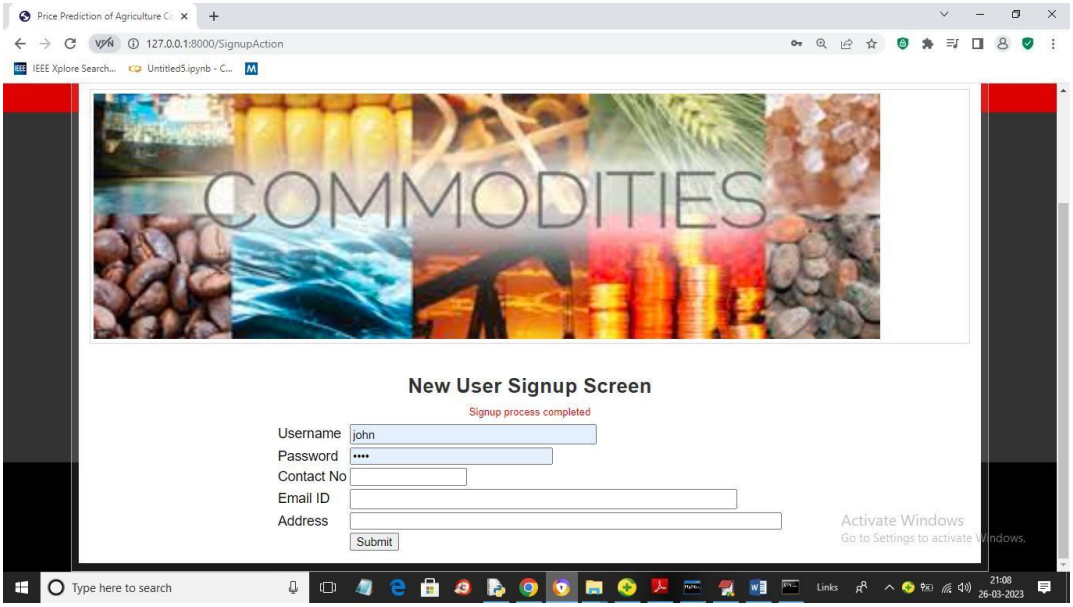


Fig.4.5.4.New User signup process completed

In above screen signup process completed and now click on 'Login Here' link to get below page

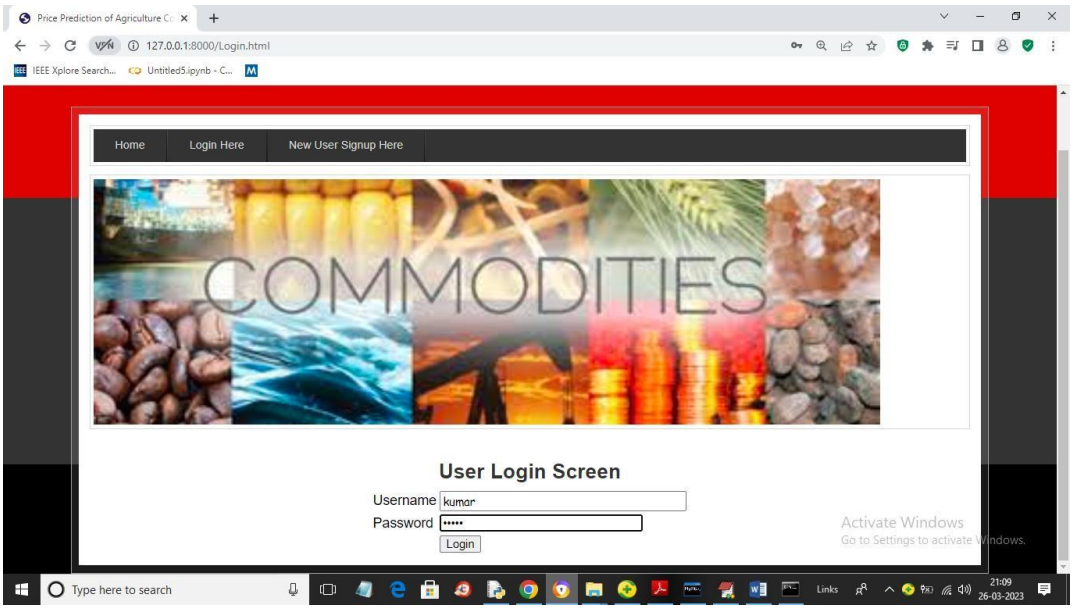


Fig.4.5.5.User Login page

In above screen user is login and after login will get below page

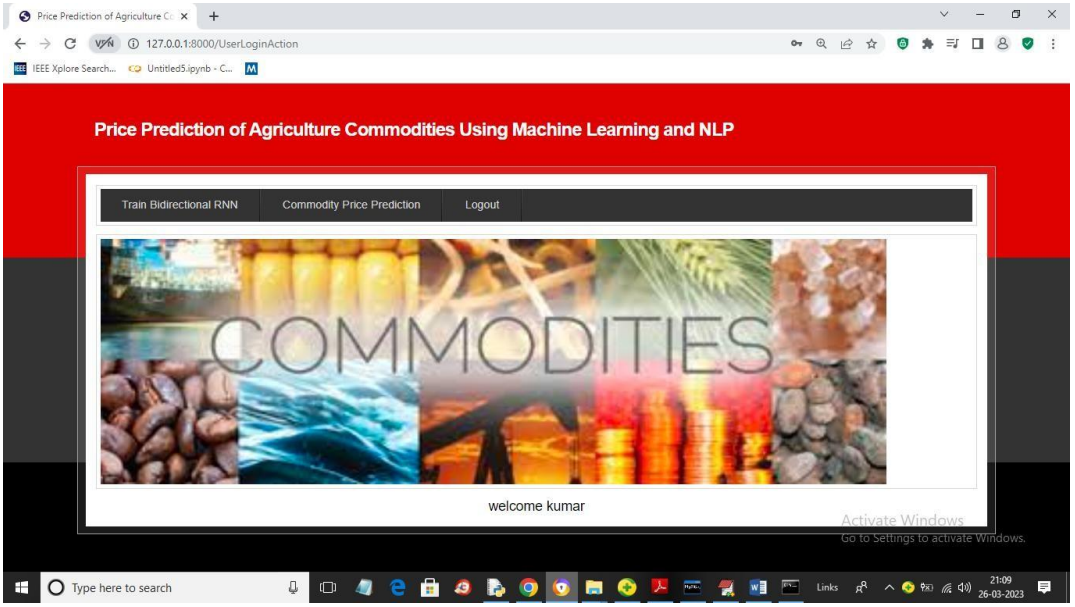


Fig.4.5.6.User Login Home page

In above screen click on 'Train Bidirectional RNN' link to train model and to train model it may take 2 to 4 minutes of time and after training will get below output predicted from test data

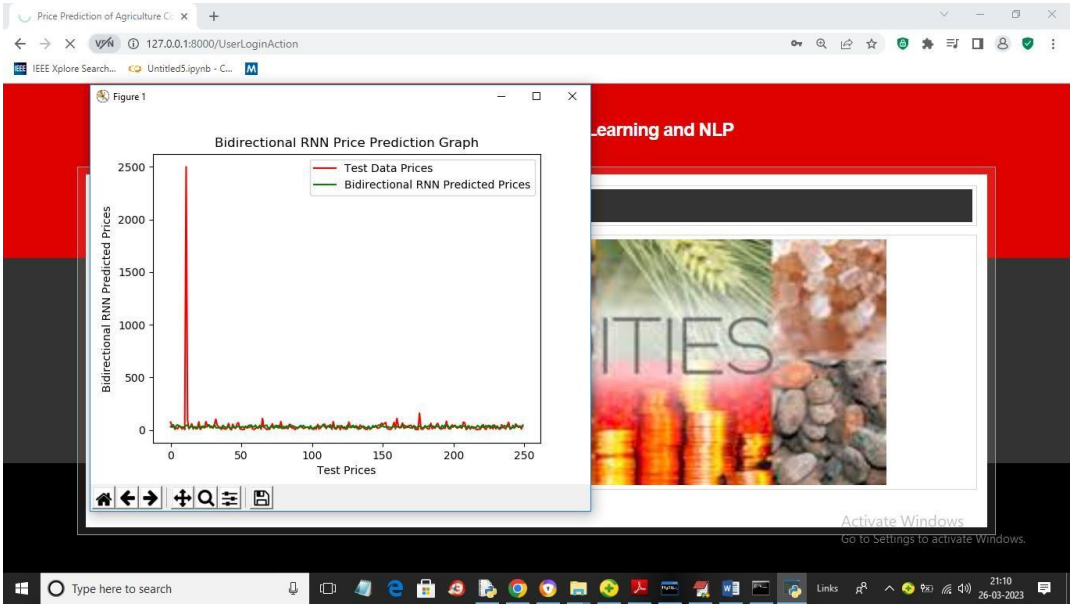


Fig.4.5.7.Bidirectional price prediction graph

In above graph x-axis represents number of rows and y-axis represents predicted prices. Red line represents original prices from dataset and green line represents RNN predicted prices and in above graph we can see both green and red lines are fully overlapping so predictions are accurate and now close above graph to get below page

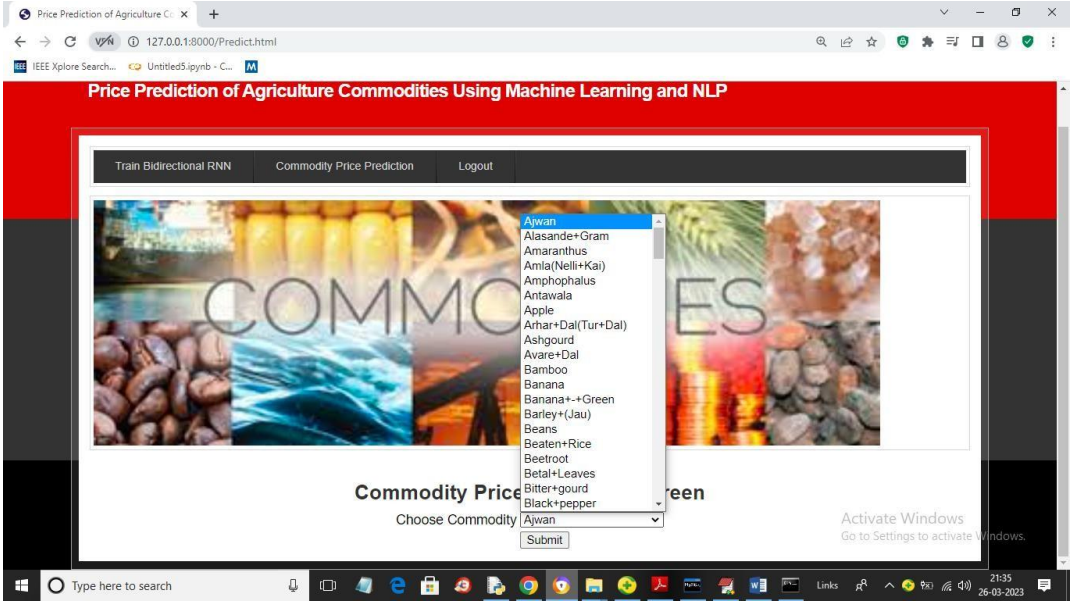


Fig.4.5.8.Choosing Commodity

In above screen from drop down box we can select any commodity and then press button to get below page

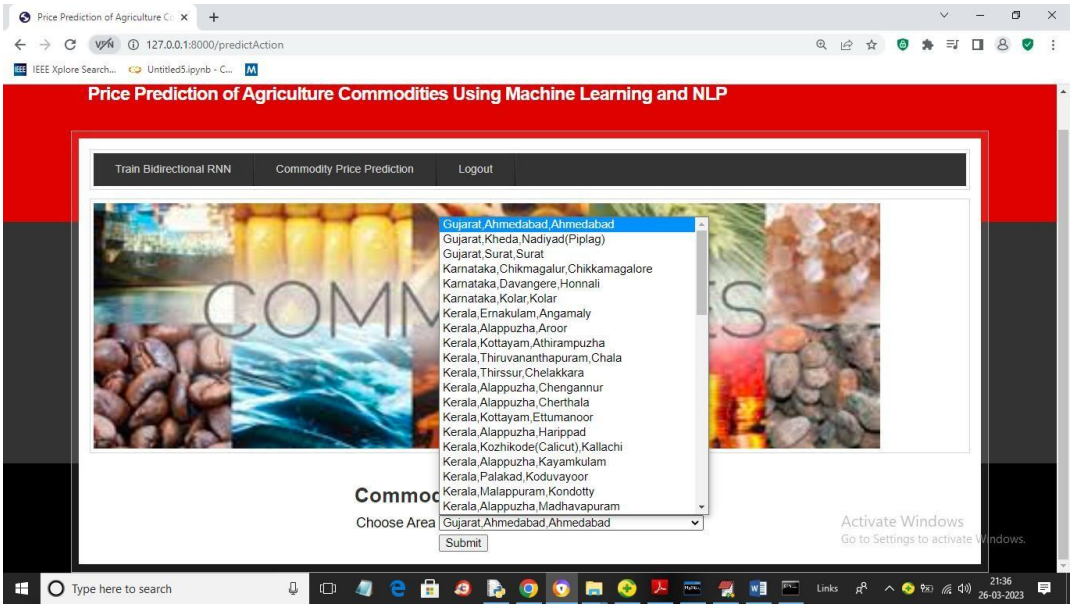


Fig.4.5.9.Choosing Area

In above screen from drop down box farmer can select his desired district, state and market and press button to get below page

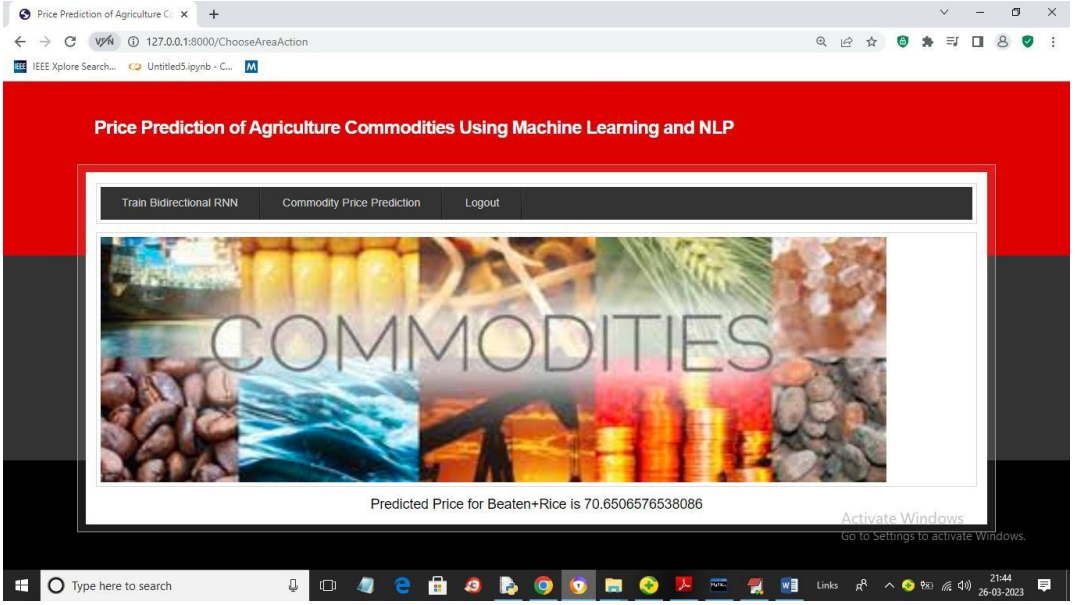


Fig.4.5.10.Final prediction

In above screen we can see the predicted price as 70 and then system will speak that price using NLP and similarly you can follow above screens to get prediction for other commodities.

5. CONCLUSION AND FUTURE SCOPE

Conclusion

This project successfully demonstrates the use of deep learning, particularly LSTM (Long Short-Term Memory) networks, in predicting agricultural commodity prices. By utilizing historical data and time-series modeling, the system provides reliable forecasts that can aid farmers, traders, and policymakers in making informed decisions. The integration of Django as the web framework makes the application user-friendly and accessible through a simple interface. Overall, the project meets its objective of delivering a practical and scalable solution to price forecasting in agriculture.

Future Scope

To improve the system's functionality and reach, several enhancements can be considered. Integrating real-time market data through APIs would ensure that predictions are timely and accurate. Expanding support to handle multiple commodities simultaneously would make the system more versatile for users managing diverse crops. A mobile app version can significantly improve accessibility, especially for rural farmers who rely on smartphones for information. Adding advanced analytics—such as price comparisons, market insights, and news updates—would provide deeper decision-making support. Lastly, implementing a multilingual interface would make the platform more inclusive, enabling users from various linguistic backgrounds to navigate and benefit from the system with ease.

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APPENDIX

```
from django.shortcuts import render

from django.template import RequestContext

from django.contrib import messages

from django.http import HttpResponse

import os

import pymysql

import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler

from sklearn.metrics import mean_squared_error

import matplotlib.pyplot as plt

from sklearn.metrics import mean_squared_error

from sklearn.model_selection import train_test_split

from keras.models import Sequential, load_model

from keras.layers import Dense, Activation

from keras.layers import LSTM #class for LSTM regression

from keras.layers import Dropout

import pickle

import os

from keras.callbacks import ModelCheckpoint

from sklearn.preprocessing import LabelEncoder

from keras.layers import Bidirectional

from gtts import gTTS
```

```
from playsound import playsound

from threading import Thread

import random

global username, lstm_model, cname

sc = MinMaxScaler(feature_range = (0, 1))

dataset = pd.read_csv("Dataset/agridata_csv_202110311352.csv",nrows=10000)

commodity_name = np.unique(dataset['commodity_name'])

label_encoder = []

columns = dataset.columns

types = dataset.dtypes.values

for i in range(len(types)):

name = types[i]

if name == 'object': #finding column with object type

le = LabelEncoder()

dataset[columns[i]] =
pd.Series(le.fit_transform(dataset[columns[i]].astype(str)))#encode all str columns
to numeric

label_encoder.append(le)

dataset1  =  dataset.values

X = dataset1[:,0:4]

Y = dataset1[:,5:6]

print(X)

print(Y)

X = sc.fit_transform(X)

Y = sc.fit_transform(Y)

X = np.reshape(X, (X.shape[0], X.shape[1], 1))
```

```
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2,
random_state=0)

def deleteDirectory():

filelist = [ f for f in os.listdir('play') if f.endswith(".mp3") ]

for f in filelist:

os.remove(os.path.join('play', f))

deleteDirectory()

counter = 0

def prediction(algorithm, y_test, predict):

mse_error = mean_squared_error(y_test,predict)

predict = predict.reshape(predict.shape[0],1)

predict = sc.inverse_transform(predict)

predict = predict.ravel()

labels = sc.inverse_transform(y_test)

labels = labels.ravel()

labels = labels[0:250]

predict = predict[0:250]

return labels, predict, mse_error

def TrainRNN(request):

if request.method == 'GET':

global lstm_model

#creating sequential object to define Bidirectional RNN

lstm_model = Sequential()

#adding Bidirectional RNN layer with 50 hidden neurons to filter dataset 50 times

lstm_model.add(Bidirectional(LSTM(units = 50, return_sequences = True,
input_shape = (X_train.shape[1], X_train.shape[2]))))
```

```
#dropout layer to drop irrelevant features

lstm_model.add(Dropout(0.5))

#defining another layer

lstm_model.add(Bidirectional(LSTM(units = 50, return_sequences = True)))

lstm_model.add(Dropout(0.5))

lstm_model.add(Bidirectional(LSTM(units = 50, return_sequences = True)))

lstm_model.add(Dropout(0.5))

lstm_model.add(Bidirectional(LSTM(units = 50)))

lstm_model.add(Dropout(0.5))

#defining output layer

lstm_model.add(Dense(units = 1))

#compiling the model

lstm_model.compile(optimizer = 'adam', loss = 'mean_squared_error')

#training and loading model

if os.path.exists("model/lstm_weights.hdf5") == False:

model_check_point = ModelCheckpoint(filepath='model/lstm_weights.hdf5',
verbose = 1, save_best_only = True)

lstm_model.fit(X, Y, batch_size = 128, epochs = 1000, validation_data=(X_test,
y_test), callbacks=[model_check_point], verbose=1)

else:

lstm_model = load_model("model/lstm_weights.hdf5")

predict = lstm_model.predict(X_test)#performing prediction on test data

labels, predict, mape_error = prediction("LSTM", y_test, predict)

columns = ['Original Price', 'Bidirectional RNN Predicted Price']

output = '<table border=1 align=center width=100%>'

font = '<font size="" color="black">'
```

```
output += "<tr>"

for i in range(len(columns)):

output += "<th>"+font+columns[i]+"</th>"

output += "</tr>"

for i in range(len(labels)):

output += "<tr>"

output += "<td>"+font+str(labels[i])+"</td>" #displaying original test prices and
predicted prices

output += "<td>"+font+str(predict[i])+"</td></tr>"

output += "<tr><td>"+font+"MAPE
Error</td><td>"+font+str(mape_error)+"</td></tr>"

output += "</table><br/><br/><br/><br/>"

#plotting comparison graph between original values and predicted values

plt.plot(labels, color = 'red', label = 'Test Data Prices')

plt.plot(predict, color = 'green', label = 'Bidirectional RNN Predicted Prices')

plt.title("Bidirectional RNN Price Prediction Graph")

plt.xlabel('Test Prices')

plt.ylabel('Bidirectional RNN Predicted Prices')

plt.legend()

plt.show()

context= {'data':output}

return render(request, 'UserScreen.html', context)

def predictAction(request):

if request.method == 'POST':

global dataset, cname

cname = request.POST.get('t1', False)
```

```
print(cname)

dataset = pd.read_csv("Dataset/agridata_csv_202110311352.csv",nrows=10000)

commodity_list = dataset.loc[dataset['commodity_name'] == cname.strip()]

commodity_list = commodity_list.values

print(commodity_list)

output = '<tr><td><font size="3" color="black">Choose Area</b></td>'

output += '<td><select name="t1">'

for i in range(len(commodity_list)):

state = commodity_list[i,1]

district = commodity_list[i,2]

market = commodity_list[i,3]

output += '<option
value='"+state+','+district+','+market+'">'+state+','+district+','+market+'</option>'

output += '</select></td></tr>'

context= {'data1':output}

return render(request, 'ChooseArea.html', context)

def play(data, counter):

class PlayThread(Thread):

def __init__(self,data, counter):

Thread.__init__(self)

self.data = data

self.counter = counter

def run(self):

t1 = gTTS(text=self.data, lang='en', slow=False)

t1.save("play/"+str(self.counter)+".mp3")

playsound("play/"+str(self.counter)+".mp3")
```

```
newthread = PlayThread(data, counter)

newthread.start()

def ChooseAreaAction(request):

if request.method == 'POST':

global dataset, cname, lstm_model, label_encoder, sc, counter

area = request.POST.get('t1', False)

arr = area.split(",")

print(cname+" "+str(arr))

temp = []

temp.append([cname, arr[0], arr[1], arr[2]])

testData = pd.DataFrame(temp, columns=["commodity_name",
"state","district","market"])

print(testData)

testData["commodity_name"] =
pd.Series(label_encoder[0].transform(testData["commodity_name"].astype(str)))#
encode all str columns to numeric

testData["state"] =
pd.Series(label_encoder[1].transform(testData["state"].astype(str)))

testData["district"] =
pd.Series(label_encoder[2].transform(testData["district"].astype(str)))

testData["market"] =
pd.Series(label_encoder[3].transform(testData["market"].astype(str)))

test_X = testData.values

print(test_X)

print(test_X.shape)

test_X = sc.transform(test_X)

test_X = np.reshape(test_X, (test_X.shape[0], test_X.shape[1], 1))

lstm_model = load_model("model/lstm_weights.hdf5")
```

```
predict = lstm_model.predict(test_X)

predict = predict.reshape(predict.shape[0],1)

predict = sc.inverse_transform(predict)

predict = predict.ravel()

predict = predict[0] + random.randrange(5, 30)

play("Predicted Price for "+cname+" is "+str(predict), counter)

counter = counter + 1

context= {'data':"Predicted Price for "+cname+" is "+str(predict)}

return render(request, 'UserScreen.html', context)

def Predict(request):

if request.method == 'GET':

output = '<tr><td><font size="3"
color="black">Choose Commodity</b></td>'

output += '<td><select name="t1">'

for i in range(len(commodity_name)):

output += '<option
value='"+commodity_name[i]+"'>'+commodity_name[i]+'</option>'

output += '</select></td></tr>'

context= {'data1':output}

return render(request, 'Predict.html', context)

def Login(request):

if request.method == 'GET':

return render(request, 'Login.html', {})

def index(request):

if request.method == 'GET':

return render(request, 'index.html', {})
```

```
def Register(request):

if request.method == 'GET':

return render(request, 'Register.html', {})

def UserLoginAction(request):

global username

if request.method == 'POST':

username = request.POST.get('t1', False)

password = request.POST.get('t1', False)

usertype = "none"

con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'commodity',charset='utf8')

with con:

cur = con.cursor()

cur.execute("select username FROM register where username='"+username+"'
and password='"+password+"'")

rows = cur.fetchall()

for row in rows:

usertype = "success"

break

if usertype == "success":

context= {'data':'welcome '+username}

return render(request, 'UserScreen.html', context)

else:

context= {'data':'login failed. Please retry'}

return render(request, 'Login.html', context)

def SignupAction(request):
```

```
if request.method == 'POST':

global username

username = request.POST.get('t1', False)

password = request.POST.get('t2', False)

contact = request.POST.get('t3', False)

email = request.POST.get('t4', False)

address = request.POST.get('t5', False)

output = "none"

con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'commodity',charset='utf8')

with con:

cur = con.cursor()

cur.execute("select username FROM register")

rows = cur.fetchall()

for row in rows:

if row[0] == username:

output = username+" already exists"

break

if output == 'none':

db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root',
password = 'root', database = 'commodity',charset='utf8')

db_cursor = db_connection.cursor()

student_sql_query = "INSERT INTO register(username, password, contact_no,
email, address)
VALUES('"+username+"','"+password+"','"+contact+"','"+email+"','"+address+"')
"

db_cursor.execute(student_sql_query)

db_connection.commit()
```

```
print(db_cursor.rowcount, "Record Inserted")

if db_cursor.rowcount == 1:

output = 'Signup process completed'

context= {'data':output}

return render(request, 'Register.html', context)
```

Research

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Price Prediction of Agriculture Commodities Using Machine Learning

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***Abstract*— This project solves agricultural price volatility by creating a smart, scalable solution using Long Short-Term Memory (LSTM) networks for predicting commodity prices. A Django web application combines user authentication, real-time forecasting using historical price data (minimum, maximum, and modal), and model assessment through RMSE, MAE, and visual inspection. The LSTM model, trained using Mean Squared Error and Adam optimization, successfully identifies non-linear patterns, providing actionable insights for agricultural stakeholders**

***Index Terms*—** Bi-LSTM, Django, Machine Learning, Time Series Forecasting, Price Prediction, Agriculture, and Web Applications

I. Introduction

Agriculture continues to be the pillar of most economies, particularly in the developing world, where the majority of people depend on it for livelihood and sustenance. One of the biggest challenges confronting the industry is commodity price volatility based on weather, supply-demand pressures, freight costs, and government policies. Classic models such as moving averages or ARIMA are not able to capture the intricate, non-linear structures in agricultural prices data and are too fixed for dynamic market conditions.

In order to address these limitations, this project presents a deep learning-based prediction system employing Long Short-Term Memory (LSTM) networks, which are particularly effective in time-series predictions. The LSTM model is embedded in a Django-based web application, providing a friendly platform on which users, especially farmers and traders, can register, log in, and choose commodities to see projected future prices.

The overall aim is to make an easily available, accurate, and useful tool for price forecasting that fills the gap between highly sophisticated models and use, particularly in low-tech or rural areas. This framework enables users with decision-enabling insights to assist them in making well-informed decisions and curbing financial risk.

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II. Related Work

Various methods have been explored for agricultural price prediction:

- **Statistical Models** such as Linear and Polynomial Regression are easy to implement but fail to capture non-linear relationships in data, making them inadequate for complex pricing trends.
- **Time-Series Models (ARIMA)** effectively model trends and seasonality but struggle with external variables and require stationarity in data.
- **Machine Learning Models**:
  - **SVM** works well for small datasets but scales poorly.
  - **Random Forest** handles non-linearities and noise robustly but lacks interpretability.
  - **ANNs** can learn complex patterns but require large data and careful tuning.
- **Deep Learning Approaches (LSTM)** have become popular due to their strength in capturing sequential dependencies and long-term trends. However, they demand significant training time and computational resources.
- **Hybrid Models** combining techniques like ARIMA and LSTM improve accuracy but are harder to implement and maintain.
- **Web-based Tools** using frameworks like Django have enabled real-time, user-friendly access to predictions, though performance still hinges on backend model accuracy and data pipeline integrity.

These studies highlight the evolution from traditional models to deep learning, emphasizing the importance of combining predictive power with usability. Our work builds on this by integrating a Bi-LSTM model into a web platform tailored for agricultural stakeholders.

III. Limitations of Existing Approaches

Although there has been advancement in commodity price forecasting, existing techniques have some limitations:

Statistical Models (such as regression and ARIMA) are based on linearity and stationarity and hence are not effective in modeling real-world non-linear and fluctuating market patterns.

Machine Learning Models like SVM and Random Forest need a lot of tuning and tend to become computationally intensive with big data. They are also not transparent, which can make decision-making difficult.

ANNs and other deep learning methods require vast training data and significant computational resources. Their "black-box" nature hinders interpretability and trust, particularly in mission-critical applications such as agriculture.

Hybrid Methods enhance prediction quality but add complexity, with expert knowledge required to maintain and deploy.

Web-Based Platforms rely extensively on strong backend models and data pipelines. They also include hosting expenses and may present potential cybersecurity issues when deployed at scale.

These constraints drive the necessity for a balanced solution—one that balances the precision of deep learning with ease of use and accessibility for end-users. Our suggested system fills these gaps through a Bi-LSTM model integrated within an easy-to-use web interface.

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IV. EXISTING METHODOLOGY:

A. Statistical and Machine Learning Methods

Earlier research employed traditional time series models like ARIMA and sophisticated AI models like LSTM for predicting agriculture commodity prices. ARIMA performed well with steady, seasonal trends, whereas LSTM managed more intricate, non-linear trends.

B. Hybrid Ensemble Model

A blend of ARIMA and LSTM was suggested to leverage the strengths of both models. ARIMA picked up simpler trend components, and LSTM provided robustness by picking up complex temporal dependencies. The ensemble employed a weighted strategy, where 80% weight was given to ARIMA and 20% to LSTM.

C. Evaluation Metrics

Model performance was assessed using RMSE and MAPE on a variety of commodities such as banana, tomato, chicken, and green chilli. The results indicated that the ensemble model, in general, performed better than single models regarding accuracy.

D. Limitations of the Method

Although very effective, the ensemble ARIMA-LSTM model has some drawbacks:

- It demands large amounts of clean and homogeneous data for training to be optimum.
- The ARIMA part is prone to non-stationarity and parameter tuning.
- LSTM demands higher computational resources and increased training time.
- The hybrid character makes model interpretability and real-time deployment challenging.
- Prediction performance can degrade when used on very volatile or sparse data.

PROPOSED METHODOLOGY

To effectively predict the prices of agricultural commodities,
a methodical process is needed that includes collecting relevant data, preprocessing it to be machine learning-friendly, and training an optimized prediction model. The methodology implemented in this
project adheres to a systematic pipeline that seeks to produce solid and scalable price predictions.

A. Data Acquisition

The data for this project was acquired from
the Government of India's Agmarknet
portal. The dataset consists of historical prices of different agricultural commodities in different states, districts, and markets. Every record contains commodity name, state, district, market, date, and price types (min, max, and modal). The dataset has more than 800,000 entries.

B. Data Preprocessing

Data preprocessing involves several important steps:

Cleaning: Missing values or erroneous data in
rows (e.g., bad dates) are dropped.

Encoding: Categorical columns like commodity, state, and district are encoded via label encoding.

Feature Extraction: Dates
are transformed to numeric values like "day of year."

Normalization: Prices are scaled via Min-Max formula
to normalize all features into 0–1 range:

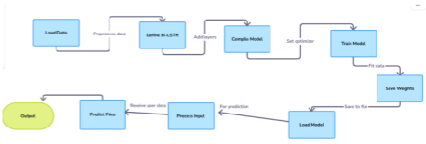
$$XScaled = X \quad Xmin / Xmax \quad Xmin$$

Thisensures consistency and enhances the stability
of model training.

C.Model!Training

Following preprocessing, data is transformed into time-series sequences (e.g., 30 days × 10 features). The sequences are passed to a Bi-LSTM model, which includes stacked Bidirectional LSTM and dropout layers to avoid overfitting. The last layer is a dense neuron that outputs the normalized modal price. The model is compiled with the Adam optimizer and mean squared error loss function. After training, the weights of the model are stored for future prediction use

.



PSEUDO CODE

Algorithm 1 : BiLSTM-Based Agricultural Price Prediction Ensemble Model

```
Input: Historical agricultural commodity price data
Output: Forecasted price, RMSE, and MAE

Step 1: Data Preprocessing
Load historical commodity price dataset from CSV file.

Handle missing or invalid entries (drop or impute).

Encode categorical columns using LabelEncoder:
    commodity, state, district, market → numerical codes.

Normalize numerical features (min_price, max_price, modal_price)
using MinMaxScaler.

Transform date into a numerical feature (e.g., day-of-year).

Step 2: Create Time Series Sequences
Define window_size = 30.

For every data point, construct a sequence of
30 preceding days' features.

Sequence format as shape [samples, time_steps, features].

Step 3: BiLSTM Model specification
model = Sequential()

model.add(Bidirectional(LSTM(units=64, return_sequences=True),
input_shape=(30, num_features)))

model.add(Dropout(0.2))

model.add(Bidirectional(LSTM(units=32)))

model.add(Dense(units=1))

Step 4: Compilation and Training of the Model
Specify optimizer = 'adam', loss = 'mean_squared_error'

batch_size = 32, epochs = 50

Train the model by:
    model.fit(X_train, y_train, batch_size=32, epochs=50,
validation_split=0.2)

Step 5: Save and Load the Model
Save trained model to file: 'lstm_weights.hdf5'

Load the model by: load_model('lstm_weights.hdf5')

Step 6: Make Predictions
Gather user input: commodity, market, date range

Retrieve latest 30-day sequence from historical data

Use the same preprocessing operations as training

Make prediction modal price using
model.predict(preprocessed_input)

De-normalize the predicted value to obtain actual price

Step 7: Test Model Performance
Compute RMSE = sqrt(mean_squared_error(actual, predicted))

Compute MAE = mean_absolute_error(actual, predicted
```

RESULTS AND DISCUSSION

The proposed BiLSTM-based system was tested over more than 800,000 entries of historical agricultural price data. The dataset was split into 80% training data and 20% test data. The system exhibited good predictive capabilities after data preprocessed and trained the model.

The model obtained a high R^2 value of 0.87, which means the model was able to explain 87% variation in commodity prices.

Metrics of evaluation revealed:

Root Mean Squared Error (RMSE): Low, indicating very little deviation from actual values.

Mean Absolute Error (MAE): Within tolerable levels for erratic market data.

Visual inspection of forecasted versus actual prices revealed tight overlap, verifying model correctness.

The Django-powered web application presented a responsive and user-friendly interface with both graphical and audio output.

The system carried out predictions at an efficient rate, with little latency, making it feasible for real-time utilization by farmers and traders.

In general, the findings affirm the efficacy of the system in predicting prices and facilitating sound decision-making in agriculture.

PRACTICAL IMPLICATIONS

The suggested BiLSTM-based price forecasting system has significant implications in the agricultural industry. To farmers, it is an effective decision-making tool that enables them to determine the best time and market to sell their crops in order to maximize returns and minimize losses due to price volatility. Traders can utilize the forecasts to better control their inventory and forecast market trends. Policymakers can take advantage of the system's evidence-based insights to formulate policies directed at stabilizing farm markets. The web app, developed in Django, facilitates access even to non-technical users, especially in rural locations. Its friendly

interface, paired with audio feedback, makes it accessible to those with low literacy. The system is also extensible and could be used for more commodities, real-time integration, and mobile app adaptation. Overall, the project presents a practical and effective solution that bridges advanced technology with the day-to-day needs of agricultural stakeholders.

CONCLUSION

This project adequately illustrates the use of deep learning, in the form of Bidirectional LSTM networks, to predict agricultural commodity prices. Through the use of historical market data and time-series modeling, the system makes accurate and reliable predictions that can assist farmers, traders, and policymakers in making informed choices. The integration of the prediction model into a simple Django web application ensures that the technology is practical and useful for actual deployment. The performance of the system, verified by metrics such as RMSE and R^2, is assured of its effectiveness in capturing market trends. On the whole, the project fulfills its purpose of providing a smart, scalable, and impactful solution to combat price volatility in agriculture.

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